MULTI-PROCESSORS

Cat Multi-Processors accept multiple interchangeable jaws for a wide range of demolition tasks. Faster cycle times get your jobs done quicker. Greater power means taking on bigger jobs. Quick jaw change gives you the right tool for the task at hand without slowing you down. All built on a durable, easy-to-maintain platform.

Features:

Cycle Times are 50 Percent Faster

- New speed booster technology dynamically shifts hydraulic force from speed to boost mode automatically as you operate.
- You'll spend less time waiting for the jaw to open or close to contact, as the speed valve automatically adjusts to fast flow when there's no load.
- Maximum crushing/cutting force is applied as soon as the jaw contacts material.

Up to 19 Percent More Force

- Process more per hour, ship more per shift, make more every day. Speed booster magnifies hydraulic force when the jaw contacts material, quickly making the cut.
- Mount more power even on smaller excavators. Compact design keeps center of gravity as close as possible to the machine.
- Get maximum performance and total support with a complete Cat demolition solution. Programs for the MP are built into the Next Gen Cat operator display. Single point of support for your entire system by your local Cat dealer.

Change Jaws in 15 minutes

- Innovative jaw locking enables a single operator to easily change jaws in 15 minutes or less. Requires only standard hand tools.
- Simple and safe jaw changes are part of the design. Every jaw sits stably, even on the roughest work site, on the included jaw stand.
- Multi-Processors use these jaw types
  - **Concrete Cutter** – outer jaw cracks concrete; inner shear jaw cuts steel
  - **Demolition** – opens wide to crack the largest concrete structures
  - **Pulverizer** – crushes concrete, separating the rebar
  - **Secondary Pulverizer** – crushes concrete into finer particles; total rebar separation
  - **Shear** – cuts structural steel, pipe, and cable
  - **Tank Shear** – cuts at right angle on three sides, cleanly cutting tanks or plate steel
  - **Universal** – innovative design cuts steel and crushes concrete

Durable and Maintainable

- Protect your investment and maintain performance, quickly and economically. Jaws are protected with easily changed, individually replaceable wear parts.
- Get more production time per cutter before replacement. Most cutting blades can be flipped in two ways to use four different cutting edges.
- Even when deep into debris, the cylinder rod is armored to prevent damage and the main cylinder body is protected inside the housing.

Multi-Processors Matching Guide:

Cat MP Model	Machine	Interface
MP318	318F, 320F, 323F, 325F, 320, 323 NGH, 340F UHD	Pin-On, CW/CPG quick Coupler and Flat top solution
MP324	326F, 329F, 330F, 335F, 336F SB, 340F UHD	Pin-On, CW/CPG quick Coupler and Flat top solution
MP332	336F, 336F SB, 340F	Pin-On, CW/CPG quick Coupler and Flat top solution
	340F UHD	Pin-On with CC or S-jaw only
MP345	349F, 352F	Pin-On, CW/CPG quick Coupler and Flat top solution
MP365	374F, 390F	Pin-On, CW/CPG quick Coupler and Flat top solution

These matches are for general reference purpose for Cat machines only.

Please always check the stability of the machine-tool configuration.

The stability depends on application, tool used and your machine configuration. For questions please contact your Cat dealer.

When choosing between various multi-processor models that can be installed onto the same machine configuration, consider work tool application, productivity requirements, and durability.

Guarding Recommendation

Multi-Processors used in hazardous applications like demolition, and scrap and material handling can create a need for special operator guarding due to flying or falling objects. When using a Multi-Processor, additional protective devices such as a front screen, Falling Object Guarding System (FOGS, includes top and front guarding), thick polycarbonate windshields or a combination of these is recommended by Caterpillar. Contact your Cat dealer for operator guarding options on your machine.

CONTRACTOR'S GRAPPLES

Cat Contractors' Grapples are a proven product built for your high-volume material handling needs. Designed for long-lasting durability, Contractors' Grapples are made to help you in the toughest of applications.

Not all attachments are available in all regions. Consult your Cat dealer for specific attachments available in your region.

Features:

Strong and Durable

- Wide surface area for demolition and material handling applications with a rigid, box-type design.
- Penetrate, securely grasp, and retain materials better with two-over-three interlocking tines.
- Keep an eye on your work with long, wide-spaced tines for better visibility.
- Tips, wrapper, and wear strips are made of AR400 steel for longlasting durability in the harshest applications.

Increase Productivity

- Debris loads and unloads seamlessly without catching due to the tips mounted flush with the wrapper.
- Optimize material flow and penetration into piles with less curvature in the lower jaw.
- Maintain high-volume production when using the grapple pinned to the machine or with a Pin Grabber Coupler.
- Purpose-designed to match their respective machine sizes for maximum performance in break out and lift capacity.
- Share your grapple across similar-sized excavators for maximum utilization with the use of replaceable pins to match different linkages.
- Capacities calculated according to SAE J2754, an industry-leading standard for fabricated grapples.
- Optimal for applications such as: structural demolition, material handling and sorting, and loading and unloading rock, scrap, pipe, waste material, and other debris.

Easy to Maintain

- No need for added hydraulics or hoses. Contractors' Grapples utilize the bucket linkages for operation.
- Long-lasting, field-serviceable tips have extended service life without extra weight.
- Jaws stay connected when the grapple pin is removed with the advanced modular pin hub system.
- Field-install an extreme duty wear package for extra protection on your grapple.

Grapple Model	G113	G117	G120	G126	G120	G126	G136	G136	G149	G174
Linkage	**312**	**315**	**B**	**B**	**CB**	**CB**	**CB**	**DB**	**TB**	**VB/VB2**
Small Excavators										
311D/F	•									
312D/E	•									
312D2/D2 GC	•									
313D2/D2 GC	•									
312F GC/313F GC	•									
312F/313F	•									
313/313 GC	•									
314D/E/F	•									
315F	•									
315/315 GC	•									
315D/316E/F		•								
318E/F		•								
318D2		•								
319D			•	•						
Medium Excavators										
320D/E/F			•	•						
320D2/D3			•	•						
320 GC			•	•						
320			•	•						
321D			•	•						
323D/D3			•	•						
323E/F			•	•						
323			•	•						
324D/E					•	•	•	•		
325D/F					•	•	•	•		
325					•	•	•	•		
326D2/F					•	•	•	•		
326					•	•	•	•		
328D					•	•	•			
329D/E/F					•	•	•	•		
330D								•	•	
330D2					•	•	•	•		
330F					•	•	•	•		
330 GC					•	•	•	•		
330					•	•	•	•		
335F					•	•	•			

• Match No Match

Contractor's Grapple Matching Guide
- Large Excavators
- Wheeled Excavators

Grapple Model	G113	G117	G120	G126	G120	G126	G136	G136	G149	G174
Linkage	**312**	**315**	**B**	**B**	**CB**	**CB**	**CB**	**DB**	**TB**	**VB/VB2**
Large Excavators										
336D/D2/E								•	•	
336D2 GC								•	•	
336D2 XE								•	•	
336E H								•		
336F								•	•	
336F XE								•	•	
336 GC								•	•	
336								•	•	
340D2								•	•	
340F								•	•	
340								•	•	
345 GC									•	
345D/349D									•	
349D2									•	
349E/F									•	
349									•	
352F									•	
352									•	
374F										•
374										•
Wheeled Excavators										
M313D		•								
M314F		•								
M315D		•								
M315D2		•								
M315F		•								
M316D		•								
M316F		•								
M317D2		•								
M317F		•								
M318D			•	•						
M318F		•								
M320D2			•	•						
M320F			•	•						
M322D			•	•						
M322D2			•	•						
M322F			•	•						

• Match No Match

Grapple Model	G113	G117	G120	G126	G120	G126	G136	G136	G149	G174
Linkage	**312**	**315**	**B**	**B**	**CB**	**CB**	**CB**	**DB**	**TB**	**VB/VB2**
Material Handlers†										
MH3022			•	•						
MH3024			•	•						
MH3026			•	•						
MH3040					•	•	•			

† When equipped with appropriate stick and bucket linkage

• Match No Match

Cat Demolition and Sorting Grapples are designed for fast, productive material handling. Capable of high-volume, production loading and precise sorting, they enhance the productivity and efficiency of your operation. The grapples can handle anything from primary and secondary demolition to recycling to get the job done.

Not all attachments are available in all regions. Consult your Cat dealer for specific configurations available in your region.

Features:

Precise Control

- Move production-sized loads with wide shell opening for material handling.
- Synchronized shells and total load control every move with the cross-mounted cylinder.
- Maintain grip on large loads or pick, sort, and place small materials with overbite stops for edge-to-edge jaw contact and prevent overbite.
- Screen dirt and other fine materials out through skeleton and perforated shells, which also give operator good visibility to the load.
- Material sorting is quick, making it easier to sort on-site and saving on tipping fees.
- Shell movement is smooth and controlled with cylinder damping.
- Integrated stop locks the rotator and keeps the shells from drifting open during transport.

Powerful Performance

- High clamping force and fast cycle times are the result of a single, large-bore cylinder design.
- Standard Load Holding Valve:
  - Maintains pressure on cylinder and load without operator constantly engaging close function, improving operator comfort and ergonomics.
  - Keeps material secure in case of hydraulic pressure loss or hose damage for added safety.
  - When active, oil flow to the grapple is blocked, which reduces fuel consumption.
- Work close to container edges and walls. Grapple shell profile has zero clearance from cutting edge against vertical walls and edges, providing access to corners in trucks, trailers, containers, bins, and 90 degree angles.
- Easy access to internal parts through large maintenance panels.
- Get the most out of your grapple with a high torque motor and longer service intervals.

Durable and Tough

- Reinforced shells made of high-strength steel are designed for the most demanding conditions and applications, such as: primary and secondary demolition, recycling, waste transfer stations, tree removal, building retaining walls, and more.
- Material fills and flows smoothly and efficiently due to countersunk bolts in the cutting edge and smooth inner profile of the shell.
- Grapple has ample rotation power to handle twisting and pulling material apart with the motor located on the outer ring.
- Increased reliability in the hydraulic system with swivel and open/close functions run independently of rotation.
- Rotate and align the grapple to pick up and grab material from any angle without moving the machine, saving wear and tear on your undercarriage.
- Operator stays safe in the cab while having the ability to strip down entire structures with the grapple.

Reliable and Easy to Maintain

- Protected rotation with high torque motor for a long service life.
- Bearings and dust seals are inside and protected from damage and debris.
- Internal dust seals provide less axial play, additionally machined surface, optimal protection against damage, and less grease consumption.
- Minimize downtime with the easy-to-replace, abrasion-resistant cutting edge for your grapple.
- Ground level access to all grease points and removable panels make grapple maintenance simple.

Options to Meet Your Needs

- Meet your specific needs with rotating and static grapples and different shell options.
- G300 and G300 GC Grapples – Rotating
  - – Rotate G300 Series grapples 360 degrees in either direction, with no limit due to the slewing ring and motor, for precise positioning.
  - – Standard models: Optimal for severe applications and frequent use.
  - – GC models: Optimal for lighter applications and occasional use.
- G200 GC Grapples – Static
  - – Non-rotating grapple intended use with Tiltrotator.
  - – Optimal for light applications and occasional use.
- Shell Types:
  - – Skeleton shell – Suited for primary and secondary demolition and other harsh applications.
  - – Closed shell – Best when used in high-volume processing and sorting applications for large, bulky materials.
  - – Perforated shell – Largest capacity and best when used in high-volume processing and sorting applications for light material densities.
- Grapples with waste handling nomenclature can handle 33-150% more material than the standard models of the same size.
- Fixed upper head models: Some models include the CW dedicated coupler hinge plate fixed to the upper head. This creates more stability with the machine due to a lower overall build-up height and less weight.

Cat Demolition and Sorting Grapple Model	Excavator Model
G212 GC	312F, 313F, 313 GC, 313, 314F, 315F CR, 315, 316F, 318F, M314F, M314, M315F, M316F, M316, M317F, M318F, M318
G213 GC	313, 316F, 318F, M314F, M314, M315F, M316F, M316, M317F, M318F, M318
G217 GC	320F, 320 GC, 320, 323F, 323, 325F CR, 325, M320F, M320, M322F
G312 GC	311F, 312D2, 312E, 312F, 313D2, 313F GC, 313F, 313 GC, 313, 314E CR, 314F, 315F CR, 315, 316F, M314F, M314, M315D2, M315F, M315
G313 GC	312D2, 312E, 312F, 313D2, 313F GC, 313F, 313 GC, 313, 314E CR, 314F, 315F CR, 315, 316E, 316F, M314F, M314, M315D2, M315F, M315, M316F, M316, M317D2, M317F, M318F, M318
G314	311F, 312D2, 312E, 312F, 313D2 GC, 313D2, 313F GC, 313F, 313 GC, 313, 314E CR
G317 GC	316E, 316F, 318D2, 318E, 318F, 320D2 GC, 320D2, 320E, 320E RR, 320F, 320 GC, 320, 323D2, 323E, 323F, 323, 325F CR, 325, M316F, M316, M317D2, M317F, M318F, M318, M320D2, M320F, M320, M322D2, M322F, M322D2 MH, M324D2 MH, MH3022, MH3024, MH3026
G318	316E, 316F, 318D2, 318E, 318F, 320D2 GC, 320D2, 320E, 320E RR, 320F, 320 GC, 320, 323D2, 323E, 323F, 323, 325F CR, 325, M316F, M317D2, M318F, M320D2, M320F, M320, M322D2, M322F, M322D2 MH, M324D2 MH, MH3022, MH3024, MH3026
G318 WH	316F, 318D2, 318E, 318F, 320D2 GC, 320D2, 320E, 320E RR, 320F, 320 GC, 320, 323D2, 323E, 323F, 323, 325F CR, M316F, M318F, M318, M320D2, M320F, M320, M322D2, M322F, M322D2 MH, M324D2 MH, MH3022, MH3024, MH3026
G324	323D2, 323E, 323F, 323, 324E, 325, 326D2, 326F, 326, 329D2, 329E, 329F, 330D2, 330F, 330 GC, 330, 335F CR, 340 UHD, 352 UHD
G324 WH	323D2, 323F, 323, 324E, 325, 326D2, 326F, 326, 329E, 330D2, 330F, 330 GC
G332	324E, 326D2, 326F, 329D2, 329E, 330D2, 330F, 330 GC, 330, 335F CR, 336D2 GC, 336D2, 336D2 XE, 336E, 336E H, 336F, 336F XE, 336 GC, 336, 340F, 340F UHD, 340, 340 UHD
G345	329D2, 329E, 330D2, 330 GC, 330, 336D2 GC, 336D2, 336D2 XE, 336E, 336E H, 336F, 336F XE, 336 GC, 336, 340D2, 340F, 340F UHD, 340, 340 UHD, 345 GC, 349D2, 349E, 349F, 349F XE, 349, 352F, 352F XE, 352, 352 UHD

Cat GSH Orange Peel Grapples expand your capabilities. With faster cycle times and increased capacities, GSH Grapples allow you to move more while spending less. A more efficient product design increases the grapple's overall productivity and reduces maintenance costs.

Not all attachments are available in all regions. Consult your Cat dealer for specific configurations available in your region.

Features:

High Performance, Less Fuel Burn

- Move more tons per hour with faster cycle times.
- Increase hydraulic flow capacity up to 160 percent with a new rotation system.
- Improve your overall fill factor up to 140-200 percent because of refined tine curvature.
- Cat Machines are pre-programmed with optimum performance settings for your grapple to maximize the pairing and efficiency of the machine and grapple.
- Reach new heights and increase your swing control. The compact height of GSH grapples extends your capabilities and is ideal for indoor applications.

Long-lasting, Quality

- Grapple life is prolonged because of lighter, force distributing cast pieces in place of welds.
- Prevent cylinders from overextending, and avert unnecessary wear on hinge points and tine tips with heavy duty, abrasion resistant upper and lower stops on the grapple's housing.
- Strength you can count on. Solid construction inner tines and tips are built of high grade steel, resisting abrasion and metal-on-metal wear. Hinge points are cast eliminating weak points on the frame.
- Increase wear life with easy to replace, cast tine tips.

Easy to Maintain

- Spend more time working with decreased maintenance time. Usage before first service interval is increased up to 250 percent and ground level grease points are safer and easier to use.
- Integral hydraulic components have been rerouted and are protected inside the tine, decreasing tension on hoses and eliminating interference with materials.
- Easy, inside-the-tine access to the hydraulics through removable panels. Panels also include dust seals to protect the critical parts inside the tines.
- Maintain a safe working environment by using the Mounting Bracket Aid which allows the bracket to stay in an upright position while installing the grapple to the machine.

Options for Specific Applications

- Meet your specific application needs with 4 or 5-tine models and optional semi-open or closed tine styles.
  - Four tines – square footprint, clean corners in rail cars and trailers. Safe grab closure for bulky goods such as car bodies.
  - Five tines – optimal retention of smaller materials.
  - Semi-open tines work best for applications where larger steel scrap or car bodies are being moved.
  - Closed tines are optimal in instances where smaller, shredded materials are the primary use.
- GSH420 and GSH520 models now come with a standard lift eye on the bottom side of the housing where a magnet can be mounted.
- Other GSH models have the option of including a lift eye when ordering the grapple.

Cat GSV Orange Peel Grapples more options to fit your needs. With faster cycle times and increased capacities, GSV Grapples allow you to move more while spending less. A more efficient product design increases the grapple's overall productivity and reduces maintenance costs. The addition of a GC model provides an economical option while delivering the same Cat quality you expect.

Not all attachments are available in all regions. Consult your Cat dealer for specific configurations available in your region.

Features:

- Continuous, bi-directional 360° hydraulic rotation.
- Heavy-duty, fully protected cylinders.
- Tines constructed of high-strength wear-resistant steel.

High Performance, Less Fuel Burn

- Move more tons per hour with faster cycle times.
- Increase hydraulic flow capacity up to 160 percent with a new rotation system.
- Improve your overall fill factor up to 140-200 percent because of refined tine curvature.
- Cat Machines are pre-programmed with optimum performance settings for your grapple to maximize the pairing and efficiency of the machine and grapple.

Long-lasting, Quality

- Grapple life is prolonged because of lighter, force distributing cast pieces in place of welds.
- Prevent cylinders from overextending, and avert unnecessary wear on hinge points and tine tips with heavy duty, abrasion resistant upper and lower stops on the grapple's housing.
- Strength you can count on. Solid construction inner tines and tips are built of high grade steel, resisting abrasion and metal-on-metal wear. Hinge points are cast eliminating weak points on the frame.
- Increase wear life with easy to replace, cast tine tips.

Easy to Maintain

- Spend more time working with decreased maintenance time. Usage before first service interval is increased up to 250 percent and ground level grease points are safer and easier to use.
- Integral hydraulic components have been rerouted, decreasing tension on hoses and eliminating interference with materials.
- Easy access to the hydraulics makes maintenance quicker, so more time can be spent moving materials.
- An optional cylinder guard is available to protect hydraulic hoses from materials.
- Maintain a safe working environment by using the Mounting Bracket Aid which allows the bracket to stay in an upright position while installing the grapple to the machine.

Options for Specific Applications

- Meet your specific needs with semi-open or closed tine style options.
  - Semi-open tines work best for applications where larger steel scrap or pipes and beams are being moved.
  - Closed tines are optimal for smaller, shredded materials.
- GC Grapples are lighter, which make them the ideal choice for recycling applications.
- Standard Grapples have been optimized for large and shredded scrap materials.

Cat OPG Model	Excavator Model
GSH420	320 GC, 323, 325, M313, M314, M318
GSH425	320 GC, 323, 325, 326, 330 GC, 330, MH3040
GSH440	326, 330 GC, 330, 336 GC, 336, 340, 340 UHD, MH3040
GSH455	336 GC, 336, 340, 340 UHD, 345 GC, 349, 352, 352 UHD
GSH520	320 GC, 323, 325, M313, M314, M318
GSH525	323, 325, 326, 330 GC, 330, MH3040
GSH555	336 GC, 336, 340, 340 UHD, 345 GC, 349, 352, 352 UHD
GSM-50	336 GC, 336, 340, 340 UHD, 345 GC, 349, 352, 352 UHD
GSM-60	345 GC, 349, 352, 352 UHD, 374D, 374F
GSV520 GC	313F, 320 GC, M313, M314, M318
GSV520	320 GC, 323, 325, M313, M314, M318
GSV525	320 GC, 323, 325, 326, 330 GC, 330

Guarding Recommendation

Orange Peel Grapples used in hazardous applications like scrap and material handling can create a need for special operator guarding due to flying objects. When using an Orange Peel Grapple, additional protective devices such as a front screen, Falling Object Guarding System (FOGS, includes top and front guarding), thick polycarbonate windshields or a combination of these is recommended by Caterpillar. Contact your Cat dealer for operator guarding options on your machine.

Selection by Application

	Grapple Application	4 Tines				5 Tines			
		O	S	C	N	O	S	C	N
Handling Scrap, Iron and Steel	Small-sized pieces (shredded)	x		■		x		■	
	Large-sized pieces up to 1000 × 1000 mm (39" × 39") (steel scrap, wrought iron, white goods, motor blocks)	■			■	■			■
	Heavy/long-sized pieces (I-beams, pipes, plates)	■		x	■			x	
	Car bodies	■	■	x	■			x	
Handling Nonferrous Scrap Metals	Small-sized pieces (beverage cans, electric devices)	x		■		x	■	■	■
	Large-sized pieces (car radiators, batteries)		■		■		■		■
	Wires and cables (copper, lead)		■		■		■		■
Other Nonferrous Materials	Waste		■		■		■		■
	Rocks, concrete blocks	■		x	■			x	■

■ Very good □ Good x Not Recommended **O** Open **S** Semi-closed **C** Closed **N** Narrow

Features:

- Multi-functional operation, the crusher combines several demolition operations in one piece of equipment. Breaking out concrete from fixed structures, pulverizing concrete and cutting reinforcement rods and small steel profiles.
- High force-to-weight ratio, the crusher's special cylinder position allows it to maintain the same power with significantly lower weight.
- Enhanced performance, the standard speed valve enables cutting/crushing with great force and in even shorter cycle times.
- Optimized serviceability, the teeth and blades are replaceable and the hydraulics is easy accessible through bolted hatches.

Guarding Recommendation

Multi-Functional Concrete Crushers used in hazardous applications like breaking out concrete from fixed structures, pulverizing concrete and cutting, can create a need for special operator guarding due to flying objects. When using a Multi-Functional Concrete Crusher, additional protective devices such as a front screen, Falling Object Guarding System (FOGS, includes top and front guarding), thick polycarbonate windshields or a combination of these is recommended by Caterpillar. Contact your Cat dealer for operator guarding options on your machine.

Matching Guide

Multi-Functional Concrete Crushers

Model	Cat Excavator
P315	315C/D, 318C, 319C/D, 320B/C/D, 322B/C, 323D, 324D, 325B/C, 325UHD C/D, 330UHD B/C/D, 345UHD B/C, 385UHD B/C
P325	320D, 322B/C, 324D, 325B/C/D, 329D, 330B/C/D, 336D, 325UHD C/D, 330UHD B/C/D, 345UHD C, 365UHD B/C, 385UHD B/C
P335	325B/C/D, 329D, 330B/C/D, 336D, 345B/C, 365UHD BII/C, 385 UHD B/C
P360	345B/C, 365B/C, 385B/C

Features:

- **Ideal for pulverizing from non fixed structures** — The Secondary Pulverizer is an excellent choice for all those recyclers who need to prepare concrete for fine crushing and is an excellent attachment for secondary demolition. Concrete chunks released during the demolition of concrete structures can be fine-crushed at source. In this process the concrete and the reinforcement are separated. This considerably reduces the transport volume, saving dumping and transportation expenses.
- **Innovative jaw arrangement** — The Cat Secondary Pulverizer offers wide jaws with pick-up tips, large opening, reversible cutting edges and fast closing times that bring a high return on your investment. This high force-to-weight work tool has been specially developed to reduce the largest possible amount of concrete in the shortest possible time.
- **Enhanced performance** — The ripper tooth splits concrete instantly and the large number of teeth have a large pulverizing effect. The cylinder of the pulverizer is equipped with a speed valve as standard. This device controls the speed of the cylinder and enables to cut/crush with great force in short cycle times.
- **Optimized serviceability** — Service and repairs are a necessary part of operating any work tool. So the less time and money needed for both, the better. That's why Caterpillar gives major attention to reducing both. The teeth and blades are replaceable and the hydraulics are easily accessible through bolted hatches. Parts commonality is designed into Cat products to significantly benefit owners of several work tools.
- **Long lifetime** — Cat Work Tools are designed to last long. Precision machined and forged parts, wear protection and stress relieving plates as well as fully protected hydraulic cylinder provide a long lifetime of low cost production.

Guarding Recommendation

Hydraulic Concrete Pulverizers are used in hazardous applications like breaking out concrete from fixed structures, pulverizing concrete and cutting; creating a need for special operator guarding due to flying objects. When using these tools, additional protective devices such as a front screen, Falling Object Guarding System (FOGS, includes top and front guarding), thick polycarbonate windshields or a combination of these is recommended by Caterpillar. Contact your Cat dealer for operator guarding options on your machine.

Matching Guide

Hydraulic Concrete Pulverizers

Model	Cat Excavator
P215	315C/D, 318C, 319C/D, 320B/C/D, 322B/C, 323D, 324D, 325B/C/D
P225	320D, 322B/C, 324D, 325B/C/D, 329D, 330B/C/D, 336D
P235	325B/C/D, 329D, 330B/C/D, 336D, 345B/C

MATERIAL HANDLERS

CONTENTS

Introduction 7-1
Our Approach and Philosophy 7-2
Broad Portfolio of
Next Generation Material Handlers 7-3
System Solutions 7-3
Ferrous Scrap Specifications 7-4

(Scrap specifications and classifications can be found in the Institute of Scrap Iron and Steel Inc.'s "Handbook." The common unit measure for the scrap industry is the gross ton which is 2240 pounds. However, short tons, net tons and metric tons may also be used.)

The versatility of Cat® material handlers, plus their ability to be equipped in any number of ways, make them an effective, low cost way to handle scrap and other materials.

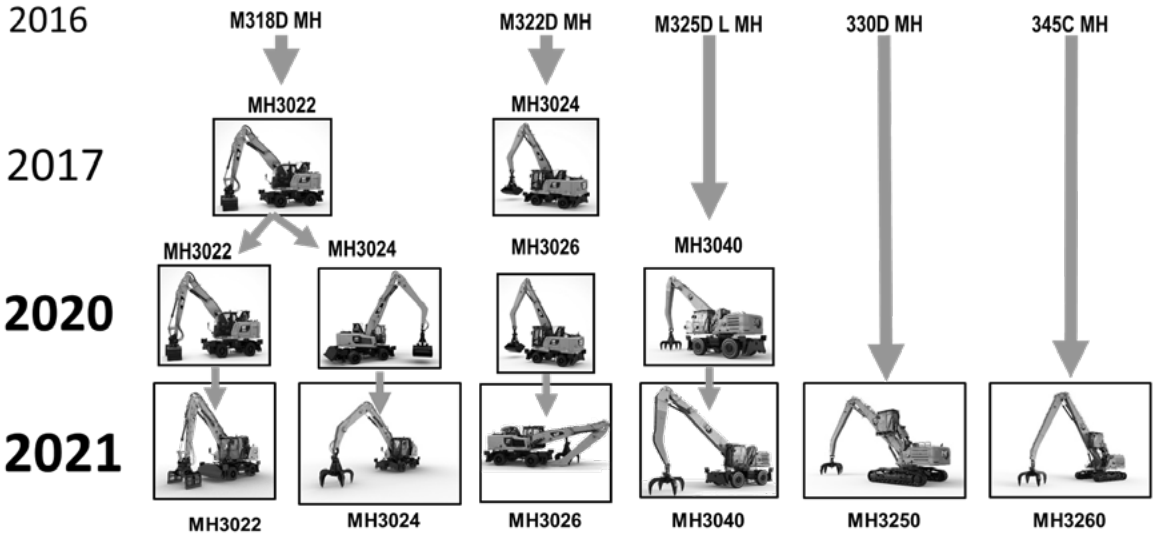
NOTE: Contact your Cat dealer for additional information on equipping Cat Material Handlers for scrap or bulk material handling applications.

Design Features

- Focused on delivering superior performance
- Sharing 100% validated Caterpillar proven components
- Driving lower owning and operating cost
- Delivering highest levels of operator comfort
- Ensuring jobsite safety – operator, technician, ground level personnel

Customer Value Message

- Reliability from proven components
- Serviceability – grouped service points, parts commonality, easy access
- Different modes (SMART, ECO, POWER) matched to the application
- Stability and ground clearance matched to underfoot conditions
- One stop shop – machine, attachment, generator



System Solutions



MH3022 MH3024 MH3026 MH3040 MH3250 MH3260

Recycling & Waste

Scrap, Forestry

Ports, Harbour

G318 and G318 WH	G324 and G324 WH		
GSH420, GSH520	GSH425, GSH525	GSH440	GSH445, GSH555

INSTITUTE OF SCRAP RECYCLING INDUSTRIES INC.

Scrap Specifications Guidelines for Ferrous Scrap

ISRI Code	Definition
200	**No. 1 heavy melting steel.** Wrought iron and/or steel scrap 6.35 mm (¼ in) and over in thickness. Individual pieces not over 1524 × 610 mm (60 × 24 in) (charging box size) prepared in a manner to insure compact charging.
203	**No. 2 heavy melting steel.** Wrought iron and steel scrap, black and galvanized, 3.18 mm (⅛ in) and over in thickness, charging box size to include material not suitable as No. 1 heavy melting steel. Prepared in a manner to insure compact charging.
207	**No. 1 busheling.** Clean steel scrap, not exceeding 305 mm (12 in) in any dimensions, including new factory busheling (for example, sheet clippings, stampings, etc.). May not include old auto body and fender stock. Free of metal coated, limed, viteous enameled, and electrical sheet containing over 0.5 percent silicon.
208	**No. 1 bundles.** New black steel sheet scrap, clippings or skeleton scrap, compressed or hand bundled, to charging box size, and weighing not less than 34 kg (75 lb) per cubic foot. (Hand bundles are tightly secured for handling with a magnet.) May include Stanley balls or mandrel wound bundles or skeleton reels, tightly secured. May include chemically detinned material. May not include old auto body or fender stock. Free of metal coated, limed, viteous enameled, and electrical sheet containing over 0.5 percent silicon.
209	**No. 2 bundles.** Old black and galvanized steel sheet scrap, hydraulically compressed to charging box size and weighing not less than 34 kg (75 lb) per cubic foot. May not include tin or lead-coated material of vitreous enameled material.
210	**Shredded Scrap.** Homogeneous iron and steel scrap magnetically separated, originating from automobiles, unprepared No. 1 and No. 2 steel, miscellaneous baling and sheet scrap. Average density 23 kg (50 lb) per cubic foot.
211	**Shredded Scrap.** Homogeneous iron and steel scrap magnetically separated, originating from automobiles, unprepared No. 1 and No. 2 steel, miscellaneous baling and sheet scrap. Average density 34 kg (70 lb) per cubic foot.
219	**Machine shop turnings.** Clean steel or wrought iron turnings, free of iron borings, nonferrous metals in a free state, scale, or excessive oil. May not include badly rusted or corroded stock.
231	**Plate and structural steel, 1.5 m (5 ft) and under.** Cut structural and plate scrap, 1.5 m (5 ft) and under. Clean open hearth steel plates, structural shapes, crop ends, shearings, or broken steel tires. Dimensions not less than 6.35 mm (¼ in) thickness, not over 1.5 m (5 ft) in length and 457 mm (18 in) in width. Phosphorous or sulphur not over 0.05 percent.
234	**Punchings and plate scrap.** Punchings or stampings, plate scrap, and bar crops containing not over 0.05 percent phosphorous or sulphur and not over 0.5 percent silicon, free from alloys. All materials cut 305 mm (12 in) and under, and with the exception of punchings or stampings, at least 3.18 mm (⅛ in) in thickness. Punchings or stampings under 152 mm (6 in) in diameter may be any gauge.

MINING & OFF-HIGHWAY TRUCKS

CONTENTS

Features 8-2
Truck Specifications 8-4
Use of Brake Performance Curves 8-5
Fixed Times for Hauling Units 8-5
Calculating Powertrain Efficiencies. 8-6

Features:

- **Cat® Four-Stroke Diesel Engines** — Turbocharged, aftercooled, adjustment-free fuel system (direct injection).
- **Electronically-Controlled Automatic Transmission** — Speed sensing device automatically shifts transmission between 1st and gear selected by operator.
- **Truck Production Management System (TPMS)** — Utilizes strut pressure sensors and an on-board microprocessor to determine payload weight, cycle segment times, delay times, actual clock time and date of each cycle.
- **Basic Health or Advanced Health (Formerly VIMS)** — Monitors all vital machine functions. Keeps operator informed of current machine operating conditions, helps reduce downtime and allows service personnel easy access to data for fast accurate diagnosis. Basic Health or Advanced Health (Formerly VIMS) includes the Truck Production Management System.
- **Mechanical Electronic Unit Injection (MEUI™)** in the 770 through 793D — Electronically maintains fuel settings, provides automatic altitude and air filter restriction compensation, and features automatic variable timing, improved diagnostics and increased fuel efficiency.
- **Common Rail Fuel System** — in the C175 engines are used on (793F, 794 AC, 795F AC, and 797F) trucks.
- **Oil Cooled Disc Brakes —** Provide retarding, service, parking, and secondary braking in a single sealed, fade-resistant, maintenance-free unit. Standard on the 770 through 775G and 777E, front brakes are caliper disc and can be switched out of the service system when not needed but activate as part of the secondary system. Standard on the 777G through 797F, front brakes are oil-cooled disc, providing excellent control in slippery conditions.
- **Automatic Retarder Control (ARC)** — Electronically controls braking on grade to maintain optimal engine RPM and oil cooling. ARC benefits include engine overspeed protection, ease of operation, faster downhill speeds, smoother ride and better control in slippery conditions.
- **Traction Control System (TCS)** — Electronically monitors and controls rear wheel slippage for greater traction and enhanced truck performance in poor underfoot conditions. If slippage exceeds a set limit, the oil-cooled disc brakes engage to slow the spinning wheel. Torque is then automatically transferred to the wheel with better traction. Available on the 770 through 797F.
- **Full Hydraulic Steering** — Functions with front suspension cylinders serving as kingpins.
- **Suspension Cylinders** — Four independent, self-contained, nitrogen/oil-pneumatic suspension cylinders absorb loading and road shocks. Wide spacing for stability.
- **Truck Bodies** — A variety of truck bodies are available to meet your application specific requirements. Those options include MSDII (mine specific design), X, HE, HP, dual slope and gateless coal bodies. The Caterpillar body program ensures the truck is configured for an optimal haulage solution.
- **Integral Roll Over Protective Structure (ROPS)** — Integral Four-Post ROPS cab standard on all models. Resiliently mounted to the main frame to reduce vibration and sound, the integral ROPS structure is designed as an extension of the truck frame. The ROPS/FOPS structure provides “five sided protection” for the operator and trainer.
- **Separate Hydraulic Systems** — Prevent cross contamination.
- **Safety** — Caterpillar continues to be proactive in developing construction and mining trucks that meet or exceed industry standards. Safety is an integral part of the machine design.

  An example of enhanced safety through the development of products, processes and solutions includes Cat Integrated Object Detection System which is an integrated camera and radar technology. More information on Object Detection is available in the technology section.

NOTE: Not all features are available on all models at this time.

Sustainability:

A variety of features improve sustainability in areas of decreasing waste, extending component life and lowering emissions levels. The 777G, 785D and 797F trucks offer Oil Renewal systems. The 777G, 785D, 793F, 797F trucks offer Continuous Rear Axle Filtration, Extended Life Filters and Extended maintenance intervals which aid in decreasing the amount of waste contributed to our environment.

- **Engines with advanced technology** contribute less emissions to the environment while maintaining fuel efficiency.
- **Advanced Surface Technology (AST)** is a replacement for hard chrome coatings on some steel parts, including Suspension and hoist Cylinder Rods.
- **Other features on 793F, 797F Mining trucks** include rear axle oil savers, front wheel sight glass, two piece cover on the final drive, ecology drains and brake wear indicators all of which help to maximize component life.

NOTE: Not all features are available on all models at this time.

Non-Dumper Offerings:

For certain applications the Caterpillar OEM Solution Group offers non-dumper arrangements for the 770G, 772G, 773G, 775G, 777G and 785D.

Non-dumper arrangements include a water truck and tractor configuration. For updates on additional offerings please contact Caterpillar OEM Solutions Group.

NOTE: Listed features may be standard on some models. Optional on others. Contact your Cat dealer for specific information.

TRUCK WEIGHT DEFINITIONS

Rated Gross Machine Weight	– Designated GMW for the machine – Optimum productivity and cost per ton – Used for performance curves and calculations
Base Machine Weight	This is the basic chassis weight which includes shipping fluids, but no attachments or options.
Attachment Weight	This is the selection of attachments, mandatory and optional, chosen for a particular truck configuration.
Body Weight	Weight of body specified in Body Type. Body Type and weight will change based on application.
Liner Weight	Liner Weight for Body Type specified. Liner Weights will change based on application.
Nominal Empty Machine Weight	Base Machine Weight plus full Fuel, Tires, Rims, Attachment Weight, Operator Weight, Body Weight, and Liner Weight.
Nominal Rated Payload	Rated Gross Machine Weight minus Nominal Empty Machine Weight.

TRUCK WEIGHT RELATIONS

Nominal Empty Machine Weight	Nominal Empty Machine Weight = Base Machine Weight + Operator Weight + Attachment Weight + Body Weight + Liner Weight.
Nominal Rated Payload	Nominal Rated Payload = Rated Gross Machine Weight – Nominal Empty Machine Weight.
Rated Gross Machine Weight	RGMW
Base Machine Weight	BMW
Nominal Empty Machine Weight	NEMW
Nominal Rated Payload	NRP

USE OF BRAKE PERFORMANCE CURVES

The speed that can be maintained when the machine is descending a grade with retarder applied can be determined from the retarder curves in this section when gross machine weight and total effective grade are known.

Select appropriate grade distance chart that covers total downhill haul; don't break haul into individual segments.

To determine brake performance: Read from gross weight down to the percent effective grade. (Effective grade equals actual % grade *minus* 1% for each 10 kg/metric ton (20 lb/U.S. ton) of rolling resistance.) From this weight-effective grade point, read horizontally to the curve with the highest obtainable speed range, then down to maximum descent speed brakes can safely handle without exceeding cooling capacity. When braking, engine RPM should be maintained at the highest possible level without overspeeding. If cooling oil overheats, reduce ground speed to allow transmission to shift to next lower speed range.

Brake Performance Curves are made in compliance with ISO 10268 and applicable to Sea Level and 32° C (90° F) temperature. Contact Factory for Application Specific Performance.

USE OF RIMPULL-SPEED-GRADEABILITY CURVES

For best results, use Caterpillar Fleet Production and Cost Analysis (FPC) to simulate cycle time, fuel burn, and production for Application Specific Performance inquiries. Contact Factory Representative for more information.

(See Wheel Tractor Scraper Section)

Total Effective Grade (or Total Resistance) is grade assistance *minus* rolling resistance.

10 kg/metric ton (20 lb/U.S. ton) = 1% adverse grade.

Example —

With a favorable grade of 20% and rolling resistance of 50 kg/metric ton (100 lb/U.S. ton), find Total Effective Grade.

(50 kg/metric ton) = 50 ÷ 10 = 5% Effective Grade
(from Rolling Resistance)
100 lb/ton = 100 ÷ 20 = 5% Effective Grade
20% (grade) – 5% (resistance) =
15% Total Effective Grade

TYPICAL FIXED TIMES FOR HAULING UNITS

Wait time, delays and operator efficiency all impact cycle time. Minimizing truck exchange time can have a significant effect on productivity.

Fixed time for hauling units include:
1. Truck load time (various with loading tool)
2. Truck maneuver in load area (Truck exchange) (Typically 0.6-0.8 min.)
3. Maneuver and dump time at dump point (Typically 1.0-1.2 min.)

Total cycle time is the combination of:
1. The above fixed time
2. Hauling time (Loaded)
3. Return time (Empty)

Example — assume load tool spots hauler with full bucket

			988F	5130B
cycle times			.60	.45
First pass	(dump time)		.10 min.	.05 min.
2 passes	(full cycle)		.70	.50
3 passes	"		1.30	.95
4 passes	"		1.90	1.40
5 passes	"		2.50	1.85
6 passes	"		3.10	2.30
7 passes	"		3.70	2.75
8 passes	"		4.30	3.20
9 passes	"		4.90	3.65
10 passes	"		5.40	4.10

NOTE: Other sizes of loading tools will have different cycle times. See Wheel Loader section for **average** cycle times for truck loading.

8

CALCULATING POWERTRAIN EFFICIENCIES

In selling against competitive OEM models, both mechanical drive and electric drive, powertrain efficiency is an important consideration. To better illustrate the advantages of Caterpillar's proven performance, grade horsepower, powertrain efficiency, and retarding horsepower should be compared to competitive OEM models.

Grade horsepower can be calculated by the following formula:

Metric

$$\text{grade HP} = \frac{\text{GMW (kg)} \times \text{TR} \times \text{Speed (km/h)}}{273.75}$$

English

$$= \frac{\text{GMW (lb)} \times \text{TR} \times \text{Speed (mph)}}{375}$$

where TR
(total resistance) = Rolling resistance + Grade resistance (expressed as a decimal)

English example

700,000 lb GMW, 2% rolling resistance, +8% actual grade at 8.2 mph would require 1530 HP

$$\frac{700{,}000 \times (.02 + .08) \times 8.2}{375} = 1530 \text{ HP}$$

Metric example

317 520 kg GMW, 2% rolling resistance, +8% actual grade at 13.2 km/h would require 1530 HP

$$\frac{317\,520 \times (.02 + .08) \times 13.2}{273.75} = 1530 \text{ HP}$$

We then calculate powertrain efficiency by dividing grade horsepower by the gross horsepower produced by the engine. Most electric drive trucks run at constant maximum horsepower while under load. Mechanical drive trucks, however, lug the engine and may produce somewhat less than maximum horsepower. Engine power curves must be utilized to determine exact horsepower produced.

Example

$$\frac{\text{1530 grade horsepower}}{\text{1800 gross engine HP}} \times 100 = 85\% \text{ powertrain efficiency}$$

Likewise, retarding horsepower being consumed by the retarding system can be calculated by the following formula:

Metric

$$\text{retarding HP} = \frac{\text{GMW (kg)} \times \text{TR} \times \text{Speed (km/h)}}{273.75}$$

English

$$= \frac{\text{GMW (lb)} \times \text{TR} \times \text{Speed (mph)}}{375}$$

where TR
(total resistance) = Rolling resistance + Grade resistance (expressed as a decimal)

English example

700,000 lb GMW, 2% rolling resistance, –8% actual grade at 14.7 mph would equate to –1646 HP

$$\frac{700{,}000 \times (.02 - .08) \times 14.7}{375} = 1646 \text{ HP}$$

Metric example

317 520 kg GMW, 2% rolling resistance, –8% actual grade at 23.6 km/h would equate to –1646 HP

$$\frac{317\,520 \times (.02 - .08) \times 23.6}{273.75} = 1646 \text{ HP}$$

This formula is intended for use in determining horsepower being consumed in the field based on field measurements. It is not intended to indicate how fast trucks should be operated on grade. Only job conditions, proper operating procedure, and good judgement should determine safe operating speeds during retarder use.